

Yuta Namekawa

Address: Tokyo, Japan, Email: namekawa.y.55ef@m.isct.ac.jp

EDUCATION

Institute of Science Tokyo (Tokyo, Japan)

Master of Engineering in Materials Science and Engineering

Anticipated: March 2027

GPA (4.5 scale): 3.62 (91.2%)

Institute of Science Tokyo (Tokyo, Japan)

Bachelor of Engineering in Materials Science and Engineering

April 2021 - March 2025

GPA (4.5 scale): 3.55 (90.5%)

Georgia Institute of Technology (Atlanta, GA, US)

Exchange program as an undergraduate Materials Science and Engineering student.

August 2024 - December 2024

GPA (4.0 scale): 3.75

RESEARCH EXPERIENCE

Institute of Science Tokyo (Tokyo, Japan)

October 2023 – Present

Graduate & Undergraduate Researcher, under Prof. Equo Kobayashi

- Investigated effects of natural aging, cold rolling, and artificial aging on precipitation behavior and mechanical properties in 7000-series Al alloys using DSC, TEM, and HAADF-STEM.
- Evaluated mechanical and electrical properties via micro-Vickers hardness, tensile testing, and conductivity measurements (1 manuscript under review)

National Institute for Materials Science (NIMS) (Tsukuba, Japan)

September 2026 - November 2026

Research Intern, under Dr. Makoto Watanabe

- Investigating process-microstructure-property relationships and defect formation mechanisms in additive manufacturing (AM) of metallic alloys under varying laser/electron beam parameters.

Norwegian University of Science and Technology (Trondheim, Norway)

May 2026 - July 2026

Visiting Research Student, under Prof. Randi Holmestad

- Quantified precipitate number density and volume fraction in 7000-series Al alloys via TEM/EELS and simulated HAADF-STEM diffraction patterns using *recipro*.
- Analyzed nanostructures via Scanning Precession Electron Diffraction (SPED) and automated phase mapping using HyperSpy and pyxem.

Georgia Institute of Technology (Atlanta, GA, US)

August 2024 - December 2024

Undergraduate Researcher, under Prof. Josh Kacher

- Analyzed SEM/OM micrographs of Al-Au and 5000-series Al alloys to determine interparticle distance/size effects on micro-galvanic corrosion and tensile fracture (co-authored paper in *Corrosion Science*).

PUBLICATIONS

- Y. Namekawa, Y. Rhee, M. Oh, T. Kondo, T. Sannomiya, T. Rolfseng, R. Holmestad, C. D. Marioara, and E. Kobayashi, "Synergy of natural aging, cold rolling and artificial aging temperature on precipitation and strength in an Al–Zn–Mg–Cu alloy" (under review at *Materials Science and Engineering: A*).
- Y. Namekawa, J. Edge, J. Nakamura, S. Paliwal, L. Merrill, R. Katona, P. Noell, and J. Kacher, "Investigating size and proximity effects in Al-Au micro-galvanic corrosion", *Corrosion Science* 264 (2026), 113761, <https://doi.org/10.1016/j.corsci.2026.113761>

CONFERENCE PRESENTATIONS

16th International Aluminium Conference (INALCO) (Trondheim, Norway)	June 2026
▪ Oral presentation	
150th Conference of Japan Institute of Light Metals (Toyama, Japan)	May 2026
▪ Poster presentation	
5th Norwegian-Japanese seminar on aluminium alloys (Toyama, Japan)	March 2026
▪ Oral presentation	
149th Conference of Japan Institute of Light Metals (Kanagawa, Japan)	November 2025
▪ Oral presentation	
IST-DUT-Tsinghua Joint Workshop (Tokyo, Japan)	October 2025
▪ Poster presentation	

RESEARCH CAPABILITIES

Mechanical evaluation:	Micro Vickers hardness test, Nanoindentation, Tensile test
Electrical evaluation:	Electrical conductivity test
Thermal analysis:	Differential scanning calorimetry (DSC)
Microstructural evaluation:	SEM, TEM, HAADF-STEM
Sample preparation:	Cutting, Grinding, Electropolishing for TEM samples
Software tools:	SolidWorks, Python, VBA, ImageJ, Digital Micrograph, Microsoft Office
Languages:	Japanese (Native), English (Fluent, C1 TOEFL iBT 5/6 (100/120))

SCHOLARSHIPS

Institute of Science Tokyo Fund “Proterial Scholarship”	April 2025 - March 2027
Satomi Scholarship by Satomi Scholarship Foundation	April 2025 - March 2027
Scandinavian-Japanese Sasakawa Foundation Scholarship	May 2026 - July 2026
Gyomu Super Japan Dream Foundation	August 2024 - December 2024

HONORS & AWARDS

Best Poster Presentation Award, Master’s Thesis Mid-Term Defense, Metallurgy Division, Department of Materials Science and Engineering, Institute of Science Tokyo	August 2026
Best Poster Presentation Award, 150th Conference of Japan Institute of Light Metals	May 2026
The Japan Institute of Metals and Materials & The Iron and Steel Institute of Japan – Young Student Award	March 2025
Early admission to Senior-year research laboratory (Top 10% of Department)	October 2023

INTERNSHIP EXPERIENCE

3D Architech Inc. (Cambridge, MA, USA Sendai, Japan)	February 2025 - Present
<i>Remote Engineering Intern – Heat Sink Design</i>	
▪ Conducting thermal-fluid simulations using SolidWorks Flow Simulation to optimize heat sink performance for electronic devices.	
▪ Collaborating remotely with the engineering team, enhancing skills in simulation, design iteration, and virtual communication.	

TEACHING EXPERIENCE

Institute of Science Tokyo (Tokyo, Japan)

December 2025 – January 2026

Teaching assistant, Undergraduate Laboratory Course (Year 2)

- Assisted second-year undergraduate laboratory classes in materials science and engineering.
- Supported students with experimental procedures, data analysis, and safety management.

EXTRACURRICULAR ACTIVITIES

Participant, NEMI Workshop: Digital TEM — Data Collection and Analysis, Trondheim, Norway **June 2026**

- Completed a 3-day hands-on workshop on automated TEM data acquisition and analysis, covering 3D electron diffraction (3D ED), 4D-STEM, and SPED using open-source tools (instamatic, PyJEM, pyxem/HyperSpy, Jana/PETS2).

Participant, Trondheim International Summer School on Aluminium Alloy Technology (ISSAAT), Trondheim, Norway **June 2025**

- Attended 34 MSc/PhD-level lectures on aluminium alloy technology (solidification, thermo-mechanical processing, microstructure characterization, modelling, age hardening, texture, forming and fracture, recycling, corrosion).

Participant, KAIST International Summer School, Daejeon, South Korea **July 2023**

- Participated in the course “Risk Society and Disaster Studies” in an English-speaking, multicultural environment, enhancing cultural awareness, communication, and adaptability through collaboration with international students.

Participant, Georgia Institute of Technology Leadership Program, Atlanta, GA, USA

March 2023

- Completed courses on leadership and teamwork, developing facilitation skills and effective group communication.